Protocol Analysis Report

Dataset: Файл данных f9ba0...

Protocol: Первичка 26.02.26 - expert run

Dataset ID: f9ba04ba-0f59-4dd4-828a-4f928a03bb15

Reproducibility and provenance

Run ID: run_20260302_144524_626485_685b1061
Source file: -
Source path: workspace/datasets/f9ba04ba-0f59-4dd4-828a-4f928a03bb15/source/original.raw
Import settings: sheet=-; header_row=-
Analysis dataset: rows=16, cols=4, xlsx=analysis_dataset.xlsx, parquet=analysis_dataset.parquet
Frozen analysis set: id=-; n_selected=-; mode=-; enforce=-
Reproducibility: ready=True; script=reproduce_run.py; payload=reproduce_payload.json; manifest=reproducibility_manifest.json
Bootstrap trace artifact: bootstrap_trace.json
Hypothesis discovery artifact: hypothesis_discovery.json
Verification status: passed
Steps excluded from report: 0
Protocol validation: status=failed; enabled=yes; strict=no
Protocol validation summary: checked=6; failed=2; warnings=0; global_errors=0
Validation policy: profile=focused; multiplicity=FDR (Benjamini-Hochberg); repair=FDR (Benjamini-Hochberg); reflection=yes
Multiplicity policy: correction=FDR (Benjamini-Hochberg); post_hoc=none; applied_steps=0
Bootstrap policy: enabled=yes; samples=1000; applied_steps=4
Validation artifact: workspace/datasets/f9ba04ba-0f59-4dd4-828a-4f928a03bb15/analysis/run_20260302_144524_626485_685b1061/artifacts/protocol_validation.json
Artifacts path: workspace/datasets/f9ba04ba-0f59-4dd4-828a-4f928a03bb15/analysis/run_20260302_144524_626485_685b1061/artifacts

Protocol Validation

Validation status: failed
Policy profile: focused
Validator mode: enabled=yes; strict=no
Correction policy: multiplicity=FDR (Benjamini-Hochberg); repair=FDR (Benjamini-Hochberg); reflection=yes
Multiplicity policy: correction=FDR (Benjamini-Hochberg); post_hoc=none; applied_steps=0
Bootstrap policy: enabled=yes; samples=1000; applied_steps=4
Step summary: total=6; checked=6; failed=2; warnings=0; global_errors=0
Validation findings by step:

- step_5 | mann_whitney | failed | errors: Numeric columns required, but non-numeric detected: Донор 1 (без геля) - креатинин.
- step_6 | mann_whitney | failed | errors: Numeric columns required, but non-numeric detected: col_1.

Methods

paired wide: n_steps=2; steps=step_3, step_4; targets=-

Hypothesis discovery and traceability

Mode: focused; Design: cross_sectional; Total: 3; Covered by steps: 0
Supported: 0; Not supported: 0; Not evaluated: 3

- Оценить связь col_1 и Донор 1 (без геля) - креатинин
  H0: col_1 и Донор 1 (без геля) - креатинин независимы.
  H1: Между col_1 и Донор 1 (без геля) - креатинин есть ассоциация.

Method: chi_square / fisher_exact; steps: -
Verdict: not evaluated; evidence: -

- Protocol step step_5: differences in Донор 1 (без геля) - креатинин by Донор 2 (без геля) - креатинин
  H0: Донор 1 (без геля) - креатинин does not differ across Донор 2 (без геля) - креатинин levels.
  H1: Донор 1 (без геля) - креатинин differs across Донор 2 (без геля) - креатинин levels.
  Method: mann_whitney; steps: -
  Verdict: not evaluated; evidence: -
- Protocol step step_6: differences in col_1 by col_6
  H0: col_1 does not differ across col_6 levels.
  H1: col_1 differs across col_6 levels.
  Method: mann_whitney; steps: -
  Verdict: not evaluated; evidence: -

Results

Step: Сравнение контроль vs опыт для Донора 1

ID: step_3

Type:

Method: paired wide

p-value: 0.0324

Statistic: 8.500

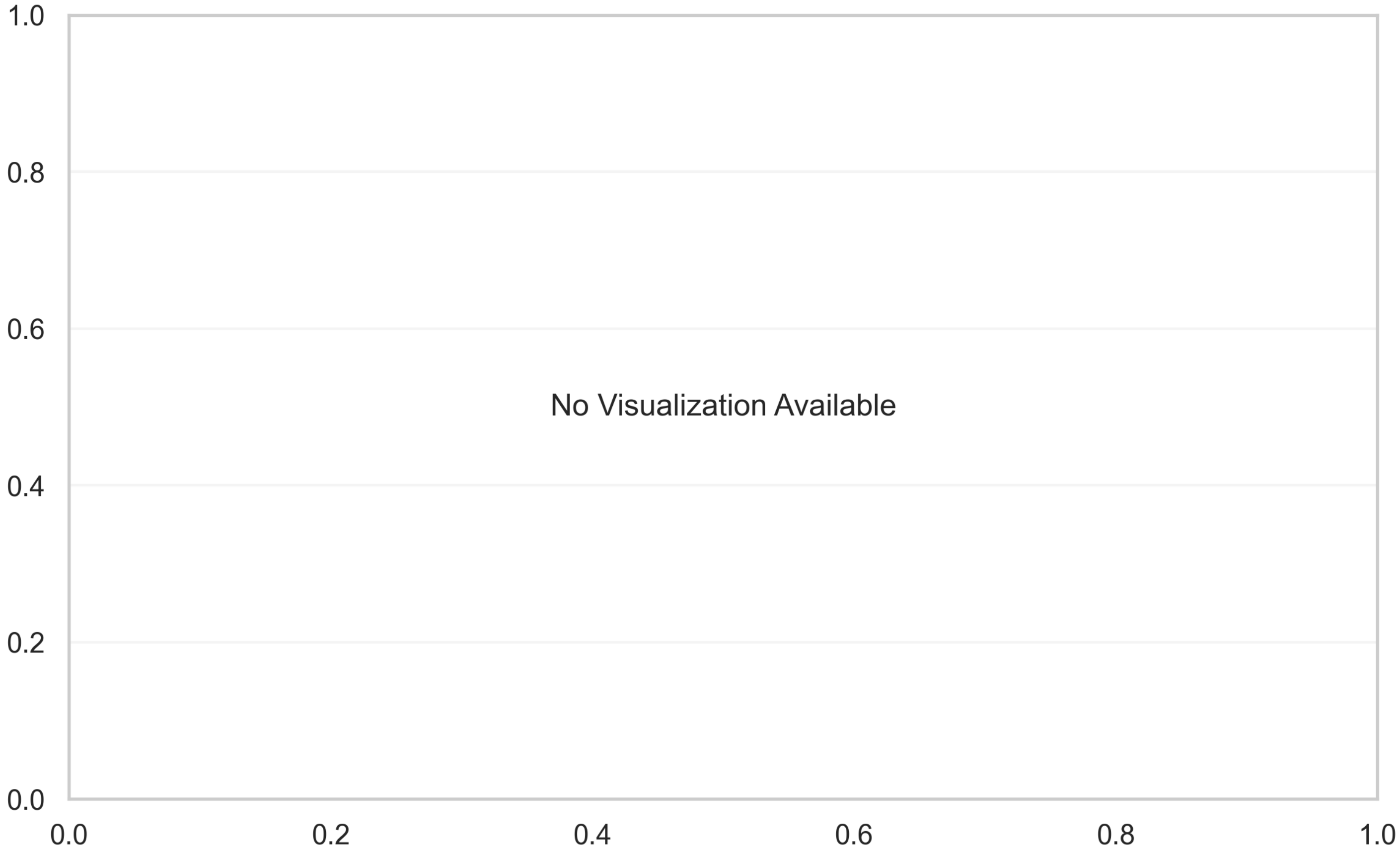
Bootstrap trace

- Bootstrap: method=-; samples=1000; ci_level=0.95
- mean_diff: est=1.326; CI [0.408, 2.250]; n_valid=1000
- effect_size: est=0.875; CI [0.241, 1.900]; n_valid=1000

Effect

Effect CI: [0.24, 1.90]

Plot: next page


No Visualization Available
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

Step: Сравнение контроль vs опыт для Донора 2

ID: step_4

Type:

Method: paired wide

p-value: 0.0321

Statistic: 6.000

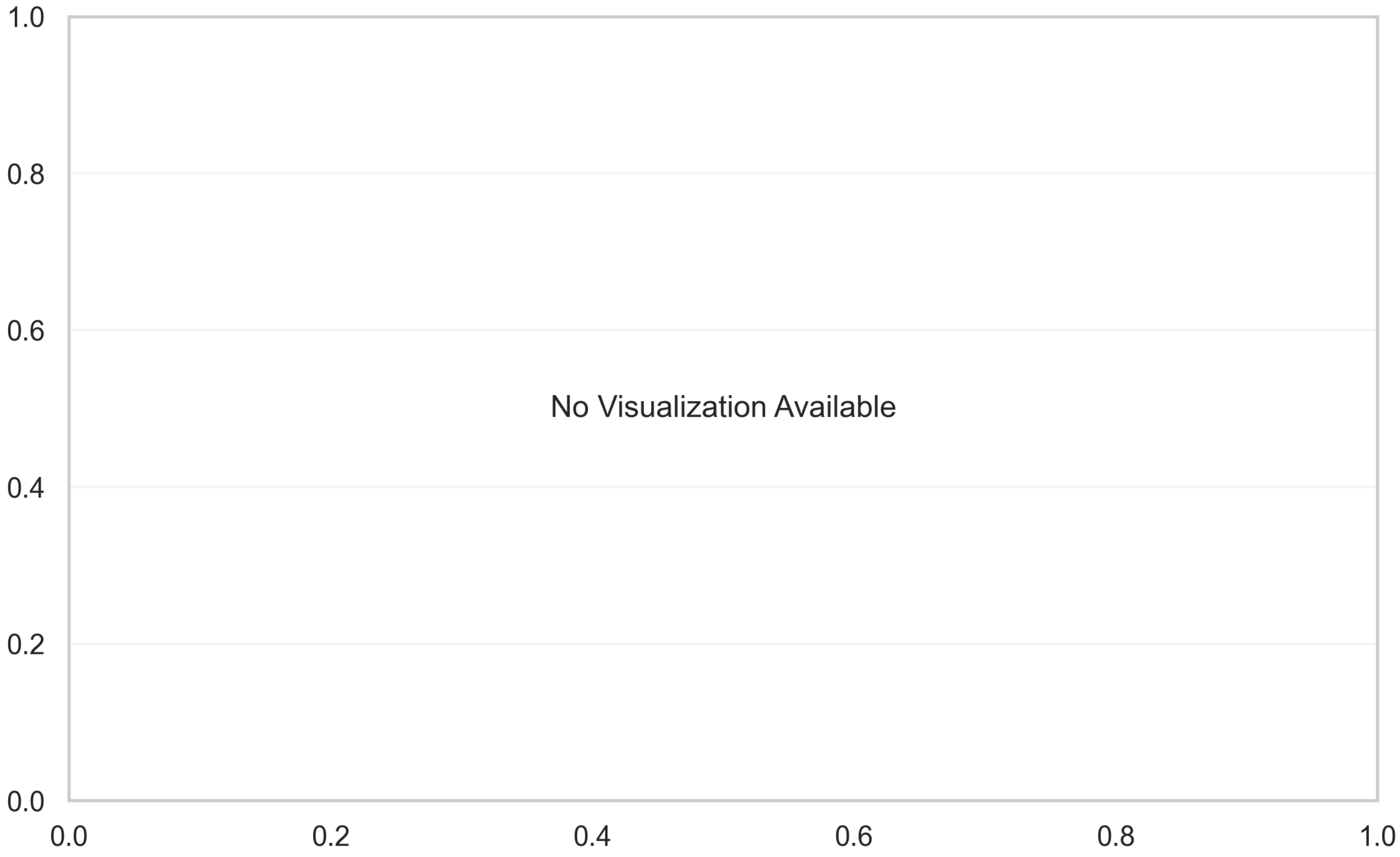
Bootstrap trace

• Bootstrap: method=-; samples=1000; ci_level=0.95
• mean_diff: est=1.067; CI [0.175, 2.017]; n_valid=1000
• effect_size: est=0.663; CI [0.125, 1.359]; n_valid=1000

Effect

Effect CI: [0.12, 1.36]

Plot: next page


No Visualization Available
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Discussion

Two hypothesis tests were conducted, both utilizing a Paired Wide method. Both tests yielded statistically significant results at an alpha level of 0.05.

The first test (step_4) showed a p-value of 0.0321 and an effect size of -0.7818. The second test (step_3) had a p-value of 0.0324 and an effect size of -0.7424.

The consistency in significance and the similar negative effect sizes across both tests suggest a robust finding.

Conclusions

- Both conducted hypothesis tests were statistically significant.
- The Paired Wide method was consistently applied.
- P-values were 0.0321 and 0.0324, respectively.
- Effect sizes were -0.7818 and -0.7424, indicating a negative effect.
- The findings are consistent across both analyses.

Limitations

- No major automatically-detected methodological limitations were found.